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Transit Innovation and Inequality: A Global Comparison

Abstract:

Transit systems have been at the forefront of modernization within the last few decades. Urban transit systems worldwide have been pressured to become more modernized within societies, but success between cities is not equal. In this paper, I explore “*Why do some cities successfully implement innovative transit systems while others lack real solutions?*” and “*How do public perception, technology, and political aspects affect these transit systems?*” London, Singapore, Tokyo, and Chicago served as comparative case studies to investigate how cultural aspects, governance models, and technological implementation determine the outcome of the transit system.

This paper starts with the Victoria Line in London as a historical example of early technological innovation that failed to maximize its potential. It failed as a result of union resistance. Singapore, alternatively, is where centralized government and public intervention enable a transit system surrounded by AI. Tokyo proposed a completely different idea, focused on the use of biomimicry to optimize transit networks, which highlights a successful idea without full implementation. Lastly, Chicago serves as a transit system to highlight the effects of a non-modern solution as a result of bureaucratic decisions, lack of funding, and safety concerns.

A focus on transit system policy that is effective, efficient, and responsive serves as a benchmark of comparison. Through the analysis of each transit system, technology alone does not determine the success or failure of modern transit systems. The success of each transit system is dependent on the government's policy making, investments in the development of modern systems, and the cultural aspect of society. Ultimately, this paper highlights that effective modern transit systems require the complete unification of technological innovations and government backing. These findings suggest that improving modern transit systems requires addressing both political and technical sides. The paper ends with recommendations for future transit systems and potential solutions.

Introduction:

Within the last few decades, cities around the world have felt the pressure of needing to have a more modern method of transportation. For example, the transit system influences many aspects of society, from economic opportunities by allowing businesses to receive new clients to environmental sustainability in having more green transportation. New technologies like automation through AI and other smart infrastructure have become available, which has led some cities to transform their transit systems into modern transportation, while other cities have fallen behind.

This paper focuses on 2 main questions that are behind the development of modern transit systems: *“Why do some cities successfully implement innovative transit systems while others lack real solutions?”* and *“How do public perception, technology, and political aspects affect these transit systems?”*. These questions are especially important for policymakers as transit is becoming more dependent on technological capabilities. This idea of more modern transportation

requires not only understanding the technological aspect of transit but also the cultural and social barriers that influence each city or country.

My argument is that technological innovations play a major role in shaping modern transit, while cultural and social aspects influence transportation as well. The success or failure of any transit system depends on how well the government can align technology, culture, and social aspects with effective policymaking. In some cities, like Singapore, this alignment has produced high-tech, efficient systems, while in others, like Chicago, political complexity and limited investment have stalled progress toward a more modern solution.

To explore these issues regarding transit systems, this paper focuses on four global cities like London, Singapore, Tokyo, and Chicago, specifically using a policy framework that evaluates the effectiveness, efficiency, and responsiveness of each transit system. First, we begin by looking at London's Victoria Line, which is considered the first automated transit system. Then we shift from a historical standpoint and focus on the main case studies in Singapore's AI-driven system, Tokyo's nature-inspired planning, and Chicago's relatively unchanged transit system. By comparing these cases, the paper reveals what drives success and what holds cities back, and can provide insights for future transit reforms.

Literature Review:

Theme 1 Technology:

One of the most central ideas when thinking about technological implementation in transit systems is the role technology plays in making the transit system more modern. Innovation like artificial intelligence(AI), automation, and digital ticketing services is a crucial technological innovation that makes the transit system more efficient for the daily user and more

responsive to urban development. As many more countries and cities around the world become more technologically reliant, transit systems need to be able to match the urban growth and development as well. Overall, within the next few years, technology is going to be a central part of many people's lives.

Singapore is often viewed as a global leader in modern technological implementation.. In particular, the country has incorporated a wide range of digital tools into its urban development efforts, especially in transportation. The Singaporean transit system is seen as a leading example of technology-driven transit. The country uses the MRT (Mass Rapid Transit) system, which focuses on using AI integration to help ensure a more optimal transit system. The FASTER (Fusion Analytics for Public Transport Emergency Response) system focuses on using data to help manage emergencies and direct traffic flow (Miller & Davenport, 2021). Similar to the FASTER system, the DART (Dynamic Autonomous Road Transit) system was used to help reduce congestion and traffic(Rau, 2019). Both systems highlight how the integration of artificial intelligence in transit systems allows countries and cities to ensure the efficiency and effectiveness of the transit system. Studies have also linked Singapore's tech-forward approach to environmental sustainability. Seldenrijch (2024), for example, argues that AI-driven transport networks have contributed to reducing carbon emissions by optimizing traffic flow and minimizing inefficiencies in the system.

Tokyo, Japan, on the other hand, focuses on a different approach to making the transit system more optimal for users. Japan is using the idea of biomimicry(the idea of using nature to help aid in urban development) by using slime models of the current Tokyo system to map out the most efficient transit system. Tero et al. (2010) found that the mold naturally grew along the shortest, most efficient paths between key nodes, mirroring much of Tokyo's real-world

infrastructure. Although Japan's system is not fully automated like Singapore's, the country's emphasis on planning through nature-inspired algorithms reflects a different but equally innovative application of technology.

On the other hand, as opposed to the modern solution present in Singapore and Tokyo, the London Victoria Line served as the first implementation of automated transit. The Victoria Line was originally launched in 1968 with a driverless transit system. However, due to the dissatisfaction of labor unions and general public concerns for safety, the system failed. This shows that technological innovation does not translate to a modern solution with social and political aspects. These examples highlight that technology is essential for a transit system to become modern. The success of each system depends on how technology is used and on political and social influences. Technological tools are only effective when paired with the infrastructure, governance, and institutional readiness to support them.

Theme 2 Government and Policy:

A second major characteristic when considering implementing more modern transit systems is the government structure and policy. Government structure and policy can determine whether technological innovations are implemented successfully or if they fail in the long run. Some cities can implement modern transit solutions more effectively than others due to the government structure. This is due to the fact of power distribution within a country, budgeting, and willingness to change an already existing system.

Singapore's government structure allows the country to rapidly develop and incorporate technology more efficiently and effectively than many other countries. This is because Singapore has a centralized government. The LTA (Land Transport Authority) is the part of the government

that focuses on transportation. This shows that centralized government authorities can make important decisions that are related to transportation more quickly and implement specific changes. Scholars like Miller (2022) and Du et al. (2023) believe that due to the centralization of government allows government officials to pass more effective policies at a quicker rate and allows for policies to be implemented faster. This allows for easier long-term planning on technological implementation between the government and the tech firms. Overall, long-term planning can take place as a result of not requiring approval from different levels of government.

On the other hand, the United States, specifically Chicago, struggles with having 3 different levels of government. The CTA (Chicago Transit Authority) has to communicate with multiple government departments and officials, as well as technological agencies, before making long-term progress in incorporating technology into transit. This highlights how long it takes for simple decisions and planning to be made. Studies by Verbas et al. (2024) and Osorio et al. (n.d.) have shown that while the Chicago government knows the transit system needs to be upgraded, there have been political debates and budgeting conflicts which has slowed down the overall urban development of a more modern system. The difference between Singapore and the United States highlights how the government structure and policy-making hinder the deployment of innovative transit solutions.

Theme 3: Public Perception and Cultural Norms:

A third factor, while considering the implementation of a more modern transit system, is the public perception and cultural norms. Although technology and public policy are a majority of what is needed to develop a modern transit system, public perception affects whether the transit system is successful or not. The people determine if the transit system is going to be used

or not. For example, the Victoria Line, which was opened in 1968, was fully automated in terms of the technology available at the time, but due to labor union resistance, the line ultimately ended up failing. This showed that technology was not enough to allow the line to succeed; the public perception of the line and willingness to accept the change also played a critical role in the failure of the line. In contrast, Singapore, which has a centralized government in which the people believe in technological innovations, allows for newer implementations in transit to be successful in the long run. The government was able to implement AI-powered systems like FASTER. Similarly, take the cultural perception that is present in Tokyo, Japan. Punctuality is a crucial part, and having reliable transportation is a crucial part. The idea of the biomimicry study (Tero et al., 2010) highlights the importance of having networks and transit systems that are optimized. On the other hand, public and cultural perception play a role in the failure of major transit systems. Take the CTA in Chicago. Concerns about crime and delays often lead transit systems not to be fully utilized. This has led to decreased ridership (Verbas et al., 2024). These examples of public perception and cultural acceptance can shape whether or not transit reforms are utilized, implemented, or delayed.

Criteria for Policy:

Modern transit systems are becoming more seen as public policy as opposed to infrastructure. Transit systems are seen as economic mobility as they allow businesses to gain clients and more consumer retention. Additionally, transit systems connect people from lower-income areas to the city. More modern solutions to transit systems allow environmentally sustainable practices as well. As technology is becoming more and more utilized in urban development and infrastructure, countries and cities are facing the pressure to change their old

transit systems into more modern up-to-date transit systems. This puts pressure on governments, and policymakers need a way to evaluate the extent to which successful transit systems are. A transit system's performance can be assessed by how well it meets public expectations.

To compare London, Singapore, Chicago, and Tokyo, we need to focus on 3 parts of the transit systems that affect transit system performance: effectiveness, efficiency, and responsiveness. These are the 3 characteristics that determine if a system can be considered to be successful or not. Additionally, these 3 characteristics are usually the most widely used when comparing transit systems. These 3 characteristics analyze how effectively technology is used in a way that allows reliable and efficient service.

When considering effectiveness, it refers to how well a specific transit system is at completing a specific goal or task. This can include reducing congestion/traffic, reducing travel times, and connecting a wider range of people within the surrounding area. An effective transit system should be able to connect people to major economic businesses and social areas. Take the Singapore FASTER and DART systems. The system is known for punctuality, the amount of time that transit arrives, and coverage (Miller & Davenport, 2021). On the other hand, the CTA struggles and can be considered ineffective. The CTA has long delays, old infrastructure, and lacks any modern changes, which leads to a less reliable network and fails to connect lower-income neighborhoods (Verbas et al., 2024). Effectiveness is ultimately how well a transit system delivers on its goals.

Efficiency is how the transit systems use the different technological implementations or modern solutions to ensure that the goals are achieved. Some of these can be how transit systems use money and infrastructure to transport people across a certain area. We can consider a system that is efficient by how well a transit system can eliminate waste and maximize productivity.

Singapore is a prime example of how a transit system is efficient. Singapore has a centralized government that integrates AI, which allows for reduced congestion and optimal train response (Rau, 2019). Additionally, Tokyo also offers an efficient solution to modern transit problems. Tokyo, although not fully automated like Singapore, has a multi-operator transit system that allows for large numbers of people to travel. Studies like biomimicry can also support the efficiency of the Tokyo transit system (Tero et al., 2010). Chicago, on the other hand, can be considered less efficient than Singapore and Tokyo. There are large costs that are associated with Chicago transit and compared to the number of people utilizing the transit system. The Chicago transit system has deteriorated over time. Efficiency in the long run is important in cities as it can show whether or not a system delivers value to the public.

Responsiveness refers to how well a transit system adapts to the changes within society, from technological innovations to societal changes. This can also include how certain routes need to change and how effective it is in an emergency. Singapore has the most responsive system. The AI integration allows the transit system to respond to real-time changes or disruptions. Similarly, Tokyo is effectively able to be as responsive in terms of the layout, which allows great crowd flow. On the other hand, the CTA has struggled to respond to the demands due to the lack of change. The Victoria Line serves as a middle ground between responsive and unresponsive. The line had effective technology but suffered from labor unions that delayed implementation.

Overall, we need to analyze the political and social aspects, not just the technological aspects. A successful transit system has all 3 of the following characteristics that meet the people's requirements and achieves its goals. Each city faces different challenges and has different strengths when it comes to its transit system. These 3 characteristics allow easy

comparison. This comparison will help reveal how technology, government decisions, and public needs all come together to shape the future of urban transit.

Cases:

Victoria Line, London:

The Victoria Line in London opened back in 1968 and was one of the first major transit systems due to automation. The Victoria Line was aimed at relieving congestion on the Underground Line. This was done through the Automatic Train Operation (ATO) that allowed trains to function automatically to a certain extent. This was done through Prime Minister Harold Wilson and the push to have a more modern system (Wolmar, 2005).

The main policy issue driving the Victoria Line was the overcrowding that the Underground line was experiencing. Additionally, the old infrastructure also caused problems due to overcrowding. The policy was aimed at creating a Victoria Line that reduces delays and congestion/overcrowding. The trains were able to run automatically without a driver, but labor unions and safety concerns led to the prevention of the full implementation of a complete automatic transit system. Automation was limited as a result, leading to trains requiring a driver (Wolmar, 2005).

Despite the problems that came after some automatic implementations, the Victoria Line improved reliability for the people and reduced travel times, leading to less overcrowding. This transit system is for the earliest implementation of automation and modernization has effective and efficient utilization. In terms of responsiveness, the system struggled to gain any traction, as many people believed that there were problems with safety and labor union restrictions. The

Victoria Line serves as a historical precedent for future automated transit systems, demonstrating both the potential and the limitations of early technological innovation in public transportation.

Singapore:

Singapore is widely known for being one of the most advanced countries. Singapore is recognized as the leading smart city. This is due to the technological implementation. The MRT (Mass Rapid Transit) system is one of the first fully technological integrations in transit. During the early 2000s, there was heavy population growth leading to urban congestion and forcing the LTA (Land Transport Authority) to find an alternative solution. This led to the development of using artificial intelligence and predictive analytics (Miller & Davenport, 2021, p. 2).

This was the major problem in managing the urban congestion while having a fully functional long-term transit system. Policy makers focused on the idea of technological innovation through digital optimization and smart systems. One of the major systems that was used was FASTER (Fusion Analytics for Public Transport Emergency Response), which monitored the railway. According to Miller and Davenport, “FASTER is designed to provide control centers with visibility over the entire rail network and recommend optimal routes in seconds” (Miller & Davenport, 2021, p. 3). This capability allows Singapore’s MRT to detect and manage services during emergencies. Additionally, Singapore also incorporated another system in the form of DART (Dynamic Autonomous Road Transit), which was focused on reducing congestion and improving connectivity between lines. Rau notes that DART “*uses a dynamic scheduling algorithm to optimize road-based services for both regular and peak-hour flows*” (Rau, 2019, p. 5). Both the FASTER and DART systems show how Singapore can

combine policy making, infrastructure, and technology to make an effective modern transit system.

Singapore's transit system performs well in terms of reducing congestion and delivering reliable service. It is considered essential to daily life in the city, supported by AI-driven systems like FASTER and DART that improve efficiency and responsiveness across the network. Similarly, the MRT system is truly efficient as there is waste reduction and less labor-intensive work. Seltenrijch explains that *"smart mobility systems in Singapore have contributed to a measurable decrease in traffic congestion and a reduction in urban carbon emissions"* (Seltenrijch 2024, p. 7). Lastly, responsiveness, the MRT is the most responsive system with centralized technology. As Miller states, *"Singapore's single-tier government makes it easier to trial new tech applications system-wide without long bureaucratic delays"* (Miller 2022, p. 2). In summary, Singapore's MRT system shows how government, public perception, and technology can all contribute to a modern transit system that is beneficial to everyone¹.

Tokyo, Japan:

Tokyo has one of the busiest transit systems in the world. The transit system carries thousands of people daily². Unlike Singapore, which focused specifically on automation and technological innovation, Tokyo focused on planning and designing. The city focused on the idea of biomimicry, which is the practice of using nature-inspired solutions to inform human systems. This focus on designing and planning helped optimize the high-demand and dense transit system.

¹ Land Transport Authority. (2023). *Public transport ridership*.
https://www.lta.gov.sg/content/ltagov/en/who_we_are/statistics_and_publications/statistics.html

² Ministry of Land, Infrastructure, Transport and Tourism. (2023). *Transport statistics annual report*.
<https://www.mlit.go.jp/en/statistics/index.html>

An example of biomimicry was using a slime mold that was modeled after the Tokyo transit system. In a study by Tero et al. (2010), researchers placed food sources on a map of Tokyo to represent key urban centers. A slime mold (*Physarum polycephalum*) naturally grew between the sources and formed a network that closely mirrored Tokyo's real subway lines. The researchers concluded that the mold "constructed networks that balance cost, efficiency, and fault tolerance, in ways that resemble human-built systems" (Tero et al., 2010, p. 2). This study became a symbol of how nature-based logic could guide infrastructure development. Other scholars, such as Becker (2011), expanded on this by applying slime mold-based algorithms to create "agent-based transport models" that allow networks to adapt dynamically (Becker, 2011, p. 5). This showed how effective the design and planning parts were for potentially incorporating a more efficient solution to a highly utilized transit system.

In terms of efficiency and effectiveness, Tokyo's current transit system achieves the goal of allowing thousands of people to travel in a very short amount of time. The transit system manages congestion fairly well. In terms of how responsive the system is, it lacks any real implementation. Using the new idea of slime molds will disrupt the current transit system, causing problems for the majority of the Japanese population. Strategic planning and network modeling can deliver a system that is just as efficient then all the current technology, like in Singapore.

Chicago:

The Chicago transit system is managed by the CTA. The current system struggles with the government making effective policy that allows the transit system to become more modern. While other cities worldwide have incorporated either technological innovations like Singapore

or the designing/planning aspect seen with the biomimicry in Japan, Chicago suffers from poor infrastructure, bad public perception due to safety concerns, and a lack of government intervention.

The major policy problem that Chicago is facing is unclear CTA coordination to have a proper solution, and a lack of funding. This is done because the CTA works with both the Chicago government and the state government, which often leads to political delays in policy. Verbas et al. (2024) describe how “multi-agency coordination problems have slowed the delivery of high-impact transit projects” and led to inefficiencies across the system (Verbas, 2024, p. 6). Although a project like the Red Line expansion has been proposed, most potential projects and infrastructure upgrades have been delayed.

In terms of effectiveness, Chicago underperforms compared to major modern systems like Singapore and Tokyo. There are constant delays with upgrades, equipment failure, and maintenance issues. Osorio et al. (2022) found that public distrust in the system worsened during the COVID-19 pandemic due to both safety concerns and inconsistent service delivery (Osorio, 2022, p. 3). In terms of efficiency, Chicago struggles since the operating costs are increasing while ridership is decreasing. Historical funding gaps have left little room for large-scale technology adoption (Kirschbaum, 1970, p. 12). Similarly, responsiveness is another area in which the CTA struggles. There have yet to be any major changes the infrastructure within Chicago. Overall, Chicago’s transit system is unable to make any significant progress towards a more modern transit system compared to other countries.

Conclusion/Comparison:

This paper looked at how four major cities in London, Singapore, Tokyo, and Chicago have tried to improve their public transit systems. Each city used different methods, and each faced different challenges. To compare them fairly, I focused on three things: how well each system works (effectiveness), how well it uses its resources (efficiency), and how easily it can adapt or improve (responsiveness).

Singapore was the most powerful overall. Its MRT is extremely reliable, efficient, and well-designed. The use of artificial intelligence, along with a strong and centralized government, enables the system to respond quickly to problems and to riders' needs. Tokyo is another example, though one that focuses more on longer-term planning and less on high-tech automation. The use of biomimicry, like the slime mold model, enables Tokyo to develop a system that is extremely organized and able to deal with massive crowds.

London's Victoria Line led the way into the modern era with its automatic train operation in the 1960s. Due to union resistance and public anxiety about safety, the line never succeeded. While the system helped to alleviate congestion, it is a prime example of how technology does not succeed unless permitted by people and politics. Chicago had the worst problems. Its transportation system is beset by delays, safety problems, and sluggish modernization. Overall, each system has its weaknesses and strengths that can be used to ensure that there is greater transit improvements in the future.

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